

osmnx-sandbox

November 17, 2025

```
[6]: %reload_ext autoreload
      %autoreload 2
```

```
[7]: from kret_studies import *
      from kret_studies.notebook import *
      from kret_studies.complex import *

      logger = get_notebook_logger()
```

```
[8]: from waymo_agent import *
```

```
[9]: G = ox.load_graphml(filepath=MAP_DIR / MANHATTAN_RAW_GRAPH)
```

```
[10]: # fig, ax = ox.plot_graph(G, node_size=0, edge_linewidth=0.5)
```

```
[11]: # 2. Define mappings for Colors AND Widths
      config = {
          "motorway": {"color": "#ff0000", "width": 3.0}, # Red, Thickest
          "motorway_link": {"color": "#ff0000", "width": 3.0},
          "trunk": {"color": "#ff6600", "width": 2.5}, # Orange, Thick
          "primary": {"color": "#ffac12", "width": 2.0}, # Gold, Medium-Thick
          "secondary": {"color": "#f4f458", "width": 1.0}, # Yellow, Medium
          "tertiary": {"color": "#ffffff", "width": 0.8}, # White, Thin
          "residential": {"color": "#555555", "width": 0.5}, # Grey, Thinnest
          "default": {"color": "#333333", "width": 0.5}, # Fallback
      }

      # 3. Generate the lists
      edge_colors = []
      edge_widths = []

      for u, v, data in G.edges(data=True):
          # Get highway type (handle lists if necessary)
          highway_type = data.get("highway", "default")
          if isinstance(highway_type, list):
              highway_type = highway_type[0]

          # Fetch style from config, or use default if type not found
```

```

style = config.get(highway_type, config["default"])

edge_colors.append(style["color"])
edge_widths.append(style["width"])

# 4. Plot with both attributes
fig, ax = ox.plot_graph(
    G,
    node_size=0,
    edge_color=edge_colors,
    edge_linewidth=edge_widths, # <--- Pass the new list here
    bgcolor="black",
    figsize=(6, 6), # Bigger size helps see the width differences
    show=True,
)

```



```
[12]: len(G.nodes), len(G.edges)
```

```
[12]: (4619, 9901)
```

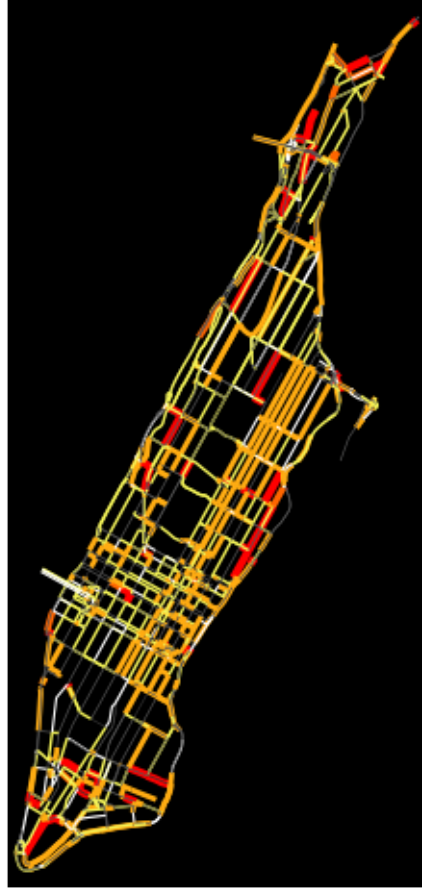
0.1 Sparsify the graph

```
[13]: G = ox.load_graphml(filepath=MAP_DIR / MANHATTAN_RAW_GRAPH)
```

```
[14]: G_sparse = sparsify_graph(G)
```

```
-----  
AssertionError                                Traceback (most recent call last)  
Cell In[14], line 1  
----> 1 G_sparse = sparsify_graph(G)  
  
File ~/coding/Columbia/RL-Project/waymo_agent/osmnx/sparsify.py:31, in  
    ↪ sparsify_graph(G)  
    29 for u, v, key, data in G_major.edges(keys=True, data=True):  
    30     highway_type = data.get("highway", "default")  
--> 31     assert isinstance(highway_type, str), f"Unexpected highway type_  
    ↪ format: {highway_type}"  
    33     if highway_type not in MAJOR_ROAD_TYPES:  
    34         edges_to_remove.append((u, v, key))  
  
AssertionError: Unexpected highway type format: ['residential', 'unclassified']
```

```
[ ]: fig, ax = ox.plot_graph(  
    G_sparse,  
    node_size=0,  
    edge_color=edge_colors,  
    edge_linewidth=edge_widths, # <--- Pass the new list here  
    bgcolor="black",  
    figsize=(6, 6), # Bigger size helps see the width differences  
    show=True,  
)  
fig.savefig(IMG_DIR / MANHATTAN_SPARSE_COLORFUL, dpi=600, bbox_inches="tight",  
    ↪ facecolor=fig.get_facecolor())
```



```
[29]: from waymo_agent import *
```

```
[ ]: from waymo_agent.osmnx.inspect import *
```

```
[31]: max_speed_distribution(G_sparse)
```

```
[31]: {'25 mph': 1125,  
      '20 mph': 203,  
      '40 mph': 62,  
      '50 mph': 35,  
      '30 mph': 60,  
      '35 mph': 11,  
      '4 mph': 2,  
      '45 mph': 21}
```

```
[ ]: inspect_graph(G)
```

Graph has 4619 nodes and 9901 edges.

Sample node attributes: {'y': 40.7443472, 'x': -73.9713203, 'street_count': 3}

```
Sample edge attributes: {'osmid': 5672217, 'highway': 'residential', 'maxspeed':  
'20 mph', 'name': 'Cooper Street', 'oneway': True, 'reversed': False, 'length':  
177.68348679429604, 'geometry': <LINESTRING (-73.924 40.868, -73.924 40.868,  
-73.923 40.868, -73.922 40.868,...>}
```

```
[61]: nx.node_connectivity(G)
```

```
[61]: 0
```

```
[ ]:
```